

Chapter 1

QBIO 305 FA26

Summary of lecture slides

Topics

- Sampling bias (ascertainment bias)
 - Some members of the population are less likely to be included
- Non-response bias
- Representative sample
- Simple random sample
 - Every member of the population has same chance of being included in the sample (representative)
 - Members are chosen independently of each other
 - It can be difficult to get a simple random sample

- Non-simple random samples can still be useful
 - Random cluster sampling
 - E.g. counting butterflies in a forest
 - Divide population into clusters
 - Take simple random sample of clusters and scale
 - Stratified random sample
 - E.g. Gallup poll
 - Divide population into groups based on characteristics
 - Take simple random samples of each group and scale

Types of evidence

- Anecdote
 - Small observation or study
- Observational study
 - Researcher collects more data as an observer
 - Conditions are not changed and no experiment is run

- Confounding variable
 - Extra variable that isn't accounted for in the study
 - Can be hard to determine
- Simpson's paradox

Case control studies

- Take a sample with the trait you're trying to study (cases)
- Take another sample without the trait
 - Controls are selected to match cases for other characteristics such as age, income, etc.

Longitudinal study (cohort study)

- Data is gathered on subjects over a period of time

Randomized control experiment

- Randomly assign patients to each group
 - Confounding variables should approximately balance between groups
- Double-blind
 - Patients shouldn't know which group they're in
 - Researchers shouldn't know which patients are getting a drug or placebo
- Researchers are doing an experiment

Sometimes randomized control experiment is not possible

- E.g. smoking, alcohol
- Sometimes sample is not the issue, it is what we're counting
 - E.g. Netflix views, crime statistics, coronavirus statistics
 - Decision making also applies to making figures